The Graded Ring

How Agent-X Decomposes Language into Algebra

Dimensions, Relations, and Projections

Agent-X Project — March 2026

The Three Dimensions

Every paragraph entering Agent-X is decomposed into a **graded ring**—an algebraic structure with one static dimension (where facts live) and two dynamic dimensions (how facts combine). The formal definition:

> A paragraph P with n sentences produces $R = \bigoplus_{i=0}^{n} R_i$, where each R_i is the set of facts from sentence i. Facts combine by addition ($\oplus$, same-kind accumulation) and multiplication ($\otimes$, cross-kind transformation), with $\otimes$ distributing over $\oplus$.

Dimension	What it does	Math	Identity	Saṅgati
Grade (sthiti)	Splits sentences into layers. Each sentence is a grade R_i. Period boundary creates new grade.	Filtration: $R = \bigoplus_i R_i$	—	sthiti
$\oplus$ **Addition** (saṅkhyā)	Accumulates same-kind quantities within/across grades. "3 apples + 2 apples = 5 apples."	Commutative group	0 (śūnya)	vṛddhi $\leftrightarrow$ kṣaya
$\otimes$ **Multiplication** (kriyā)	Transforms across kinds. "mass $\times$ velocity^2 $\times$ $\frac{1}{2}$" or "3 bags $\times$ 4 apples/bag."	Monoid	1 (eka)	spanda (impulse)

The Binding Law

Distributivity ($\otimes$ distributes over $\oplus$) is what makes this a *ring*, not just two separate operations:

$$a \otimes (b \oplus c) = (a \otimes b) \oplus (a \otimes c)$$

Example: "3 children each have 2 apples and 1 orange."
$3 \otimes (2 \oplus 1) = (3 \otimes 2) \oplus (3 \otimes 1) = 6 \oplus 3 = 9$ total fruit.
The $\otimes$ ("each"/"per") distributes over the $\oplus$ ("and" joining apples + oranges).

Grade Decomposition

The period `(.)` emits a `[``.'', ``virām'', ``.'']` triple—the grade boundary. Within a grade, comma/"and" emits `dvandva`—an intra-grade $\oplus$ separator. The question sentence (containing "find", "how many", "which") is the *projection grade*—it reads the ring, not the data.

Example: "Alice has 3 apples. Bob gave her 2 more. How many apples?"

R_0:	"Alice has 3 apples"	$\{(\text{apples}, 3)\}$
R_1:	"Bob gave her 2 more"	$\{(\text{apples}, 2), \text{dir} = \text{vṛddhi}\}$
R_2:	"How many apples?"	$\text{projection} = \text{CARDINALITY}$

Fold: $\underbrace{0}_{\text{śūnya}} \oplus \underbrace{3}_{R_0} \oplus \underbrace{2}_{R_1} = 5$

The 10 Relations (Viśeṣaṇam)

The knowledge graph is a 3D boolean tensor $G \in \{0,1\}^{N \times N \times 62}$ with 1649 nodes and 62 relation types. The **10 viśeṣaṇam** are the structural generators—every other relation is composed from them.

To see them at work, trace one question through the pipeline.

Input: "A car has mass 1000 kg and velocity 20 m/s. What is the kinetic energy?"

Phase 1: BQG — sentence $\to$ triples

Each word is resolved against the graph. The relations emitted form the raw ring elements:

Word	Triple emitted	Relation (math type)	Dim
car	[car, mithya, car]	**mithya** (rejection): car $\notin$ Graph	—
mass	[mass, satya, mass]	**satya** (assertion): mass $\in$ Graph	grade
1000	[1000, asprista-saṅkhyā, 1000]	**asprista-saṅkhyā** (unbound qty): raw v	$\oplus$
kg	[mass, saṅkhyā, 1000]	**saṅkhyā** (binding): mass $\mapsto 1000$	$\oplus$
	[mass, mātrā, kilogram]	**mātrā** (unit): mass $\in$ kg	$\oplus$
and	[and, dvandva, dvandva]	**dvandva** (compound): intra-grade $\oplus$ boundary	grade
velocity	[velocity, satya, velocity]	**satya**: velocity $\in$ Graph	grade
20 m/s	[velocity, saṅkhyā, 20]	**saṅkhyā**: velocity $\mapsto 20$	$\oplus$
.	[., virām, .]	**virām**: grade boundary $R_0 \to R_1$	grade
What	[What, vidhi-kāla, solve-for]	**vidhi-kāla** (imperative): projection signal	proj
kinetic energy	[KE, satya, KE]	**satya** + **sandhi** (compound merge)	proj
?	[?, virām, ?]	**virām**: grade boundary	grade

Phase 2: Expand — graph walk discovers the 10 viśeṣaṇam

Starting from `kinetic-energy`, the PPR walk reads edges. Each of the 10 structural relations tells the pipeline something specific:

#	Relation (saṅgati)	Math type	What the pipeline learns
1	**swarūpa** (self-nature) KE $\to$ KE	id : $x \mapsto x$	Concept exists as itself. The $\otimes$-identity: transforming by swarūpa changes nothing.
2	**abheda** (non-difference) KE $\equiv$ "$\frac{1}{2}mv^2$"	Equivalence: $x \sim y$	Alternative names/forms. KE and $\frac{1}{2}mv^2$ name the same thing.
3	**dṛṣṭhānta** (example) KE $\leftarrow$ "a moving ball"	Witness: $\exists x$	Concrete instance. Grounds the abstract concept.
4	**sthita** (foundation) KE $\leq$ energy	Partial order: $x \leq y$	Taxonomy. KE is a kind of energy. Gives grade ordering.
5	**yukta** (connection) KE — mass, velocity	Association: $x \sim y$	Which quantities are involved. The $\oplus$-summands.
6	**siddha** (established) KE $\vdash$ conservation	Axiom: $\vdash x$	What law guarantees this. Proof/verification.
7	**kriyā** (action) KE-mantra $\to$ [sq, mul, half]	Composition: $f \circ g \circ h$	The $\otimes$-chain. Operation sequence to compute result.
8	**phala** (fruit) KE-mantra $\to$ KE	Codomain: $f \to y$	Output of computation. What the krama produces.
9	**janya** (born-from) KE-mantra $\leftarrow$ mass, velocity	Domain: $x, y \to f$	Required inputs. What must be bound before computing.
10	**pratipakṣa** (opposite) half $\leftrightarrow$ double, sq $\leftrightarrow \sqrt{\ }$	Inverse: f^{-1}	How to reverse each step. Enables inverse solving.

Key pairs: phala $\leftrightarrow$ janya (output $\leftrightarrow$ input), yukta $\leftrightarrow$ rahita (connection $\leftrightarrow$ absence), kriyā $\leftrightarrow$ nirodha (action $\leftrightarrow$ cessation), abheda $\leftrightarrow$ vibheda (sameness $\leftrightarrow$ distinction).

Phase 3: Dispatch — krama evaluation ($\otimes$ chain)

The kriyā relation gave us the krama: `square` $\to$ `mul` $\to$ `half`. Each step is a $\otimes$ operation that transforms the accumulator:

Step	Op	Input	Output	Inverse (pratipakṣa)
1	square (x^2)	velocity $= 20$	400	$\sqrt{x}$ (square-root)
2	mul ($x \times y$)	$400 \times$ mass $= 400 \times 1000$	400000	x/y (division)
3	half ($x \times 0.5$)	400000	200000	$x \times 2$ (double)

Answer: kinetic-energy $= 200000$ J. To solve *inversely* ("KE is 200000, find velocity"), the pipeline reverses the krama and applies each pratipakṣa: double $\to$ div $\to$ sqrt.

The 6 Projections

The question grade determines how to read the ring result. Each question type is a projection function $\pi : R \to$ Answer.

Projection	Trigger words	Operation	Ring dimension
DERIVE	find, calculate, what is...given data	Evaluate krama $\otimes$-chain on bound values	$\otimes$
CARDINALITY	how many, total	$\oplus$-fold across grades, return count	$\oplus$
COMPARE	which, more, less, fastest	Build entity set, apply $\arg\max$ / $\arg\min$	$\oplus$ then viveka
LOGIC	is X a Y, does X have Y	Walk swarūpa/sthita edges, $\wedge$-chain	boolean
TYPED	does X $\in$ S, is X subset of Y	Set membership/subset predicate	set-ring
DEFINITION	what is X (no data)	Graph walk: describe node's edges	graph read

Every question follows the same pattern: **build the ring from the paragraph $\rightarrow$ apply the projection**. The ring is the same structure regardless of question type—only the readout differs.

Traced Examples

Example 1: Count Addition ($\oplus$ fold)

Q: "Tom has 5 apples. He gave away 2 apples. How many apples does Tom have?"

Grade	Content	Ring element	Direction
R_0	"Tom has 5 apples"	(apples, 5)	—
R_1	"He gave away 2 apples"	(apples, 2)	kṣaya (subtraction)
R_2	"How many apples..."	projection $=$ CARDINALITY	—

The word "gave away" resolves to kṣaya—the $\oplus$-inverse (subtraction). The fold:

$$\underbrace{0}_{\text{śūnya}} \oplus \underbrace{5}_{R_0} \ominus \underbrace{2}_{R_1} = 3$$

Relations active: **satya** (apples $\in$ Graph), **saṅkhyā** (5, 2), **virām** ($R_0 \rightarrow R_1$), **vidhi-kāla** (solve-for), **prati-pakṣa** (vṛddhi $\leftrightarrow$ kṣaya).

Example 2: Multiplication ($\otimes$ across grades)

Q: "Each of 3 bags has 4 apples. How many apples in total?"

Grade	Content	Ring element	Operation
R_0	"Each of 3 bags"	(bags, 3)	$\otimes$ ("each" = multiplication)
R_0	"has 4 apples"	(apples, 4)	within same grade
R_1	"How many apples..."	projection $=$ CARDINALITY	—

The word "each" triggers $\otimes$ (multiplication) instead of $\oplus$ (addition):

$$3 \otimes 4 = 12$$

This is distributivity in action: "each" means the 3 distributes over the 4. Compare with "3 apples and 4 oranges" where "and" gives $3 \oplus 4 = 7$ ($\oplus$, same-grade accumulation).

Example 3: Comparison (viveka projection)

Q: "Ball-A has mass 5 kg. Ball-B has mass 3 kg. Which ball is heavier?"

Grade	Content	Ring element	Entity
R_0	"Ball-A has mass 5 kg"	(mass, 5, kg)	Ball-A
R_1	"Ball-B has mass 3 kg"	(mass, 3, kg)	Ball-B
R_2	"Which ball is heavier?"	projection $=$ COMPARE	viveka-max

Each grade builds a separate entity. The question grade triggers viveka (discrimination)—a projection that selects from the entity set:

$$\arg\max\{\text{Ball-A} : 5,\ \text{Ball-B} : 3\} = \text{Ball-A}$$

"Heavier" maps to `viveka-max` via the graph. "Lighter" would map to `viveka-min` and return Ball-B. The ring accumulates entities across grades; viveka projects one answer from the set.

Example 4: Inverse Solve (reversed $\otimes$ chain)

Q: "Kinetic energy is 200 J and mass is 4 kg. Find the velocity."

Grade	Content	Ring element
R_0	"KE is 200 J and mass is 4 kg"	(KE, 200, J) $\oplus$ (mass, 4, kg)
R_1	"Find the velocity"	projection = DERIVE, target = velocity

Velocity is a **janya** (input) of KE-mantra, not the **phala** (output). The pipeline inverts the krama chain using **pratipakṣa** edges:

Step	Forward op	Inverse op	Input	Output
1	half	double	KE $= 200$	400
2	mul (mass)	div (mass)	$400/4$	100
3	square	$\sqrt{\ }$	$\sqrt{100}$	10

Answer: velocity $= 10$ m/s. No inverse formula was hardcoded—the engine reads pratipakṣa edges on each operation node and reverses the chain automatically.

The ring is the paragraph. The projection is the question.
Every answer is: build $\to$ fold $\to$ read.